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Essentials of Data Science
Laboratory - 2304102L - 2026

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1.1.1. Calculate Momentum

07:02

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:
$$p = m \times v$$

where:
 m is the mass of the object (in kilograms).
 v is the velocity of the object (in meters per second).
Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases

calculate...

Submit

1 `m = float (input())`

2 `v = float (input())`

3 `p = m*v`

4 `print ('%.2fkgm/s'%p)`

Terminal Test cases

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 $1 \leq n \leq 999$

- The input consists of a single integer n .

- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

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 Debugger

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1.1.3. Days between Two Dates

Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format `YYYY - MM - DD`.
- The second line contains the second date in the format `YYYY - MM - DD`.

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

Sample Test Cases

noOfDay...

1 from datetime import date

2 from datetime import datetime

3

4 # Input dates

5 date1 = input()

6 date2 = input()

7

8 # Convert string to date format

9 d1 = datetime.strptime(date1, "%Y-%m-%d")

10 d2 = datetime.strptime(date2, "%Y-%m-%d")

11

12 # Calculate difference

13 days = (d2 - d1).days

14

15 # Output

16 print(days)

17

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Test cases

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1.1.4. Reverse a Number

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Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

The input is a single integer.

Output Format:

The output prints a single integer, which is the reversed number.

Sample Test Cases

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1.1.5. Multiplication Table

00:40

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

Print the multiplication table for the given number in the format:

<number> x <multiplier> = <result>

Note:

The character "x" is used in the output to represent multiplication.

Refer to the visible test-cases for more clarity.

Sample Test Cases

multiplica...

Submit

1 # Input number

2 n = int(input())

3

4 # Print multiplication table

5 for i in range(1, 11):

6 print(f"{n} x {i} = {n*i}")

7

8

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1.2.1. Pass or Fail

1.2.2. Fibonacci Series

1.2.3. Pattern - 1

1.2.4. Pattern - 2

2. Practical 2

2.1. Practice Lab Assignment

2.2. Lab Assignment

1.2.1. Pass or Fail

Write a Python program that accepts the number of courses and the marks of a student in those courses.
The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n , the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Note:

- Aggregate Percentage: Average performance across all courses.

passorFa...

Submit

```
1 n = int(input())
2 marks = list(map(int, input().split()))
3
4 # Check fail condition
5 failed = False
6 for m in marks:
7     if m < 40:
8         failed = True
9         break
10
11 if failed:
12     print("Fail")
13 else:
14     avg = sum(marks) / n
15     print(f"Aggregate Percentage: {avg:.2f}")
16
17 if avg > 75:
18     print("Grade: Distinction")
19 elif avg >= 60:
20     print("Grade: First Division")
21 elif avg >= 50:
22     print("Grade: Second Division")
23 elif avg >= 40:
24     print("Grade: Third Division")
```

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2. Practical 2

2.1. Practice Lab Assignment

2.2. Lab Assignment

2.2.1. Linear search Technique

1.2.2. Fibonacci Series

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

- A single integer n representing the number of terms to generate.

Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

fibonacci...

1 def fibonacci(n):

2 # Recursive function to find Fibonacci number

3 if n == 1:

4 return 0

5 elif n == 2:

6 return 1

7 else:

8 return fibonacci(n-1) + fibonacci(n-2)

9

10 n = int(input())

11 for i in range(1, n + 1):

12 print(fibonacci(i), end=" ")

13

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2 Practical 2

2.1 Practice Lab Assignment

2.2 Lab Assignment

2.2.1 Linear search Technique

2.2.2 Captain of the Team

1.2.3. Pattern - 1

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases

rightangl...

1 a=int(input())
2 i=1
3 j=1
4 for i in range(a):
5 for j in range(i+1):
6 print("* ",end='')
7 print()

Terminal Test cases

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15. *Journal of the American Medical Association*, 2000; 283: 2686-2692.

- The input is an integer, representing the number of rows in the pattern.

- The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

- Refer to the displayed test cases for the sample pattern.

 $+$

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```
1 a=int(input())
2
3 k=0
4 for i in range(a):
5     for j in range(i+1):
6     k=k+1
7     print(k,"",end='')
8     k=k+1
9     print()
```